Diploma in Project Management

What is Project Management?

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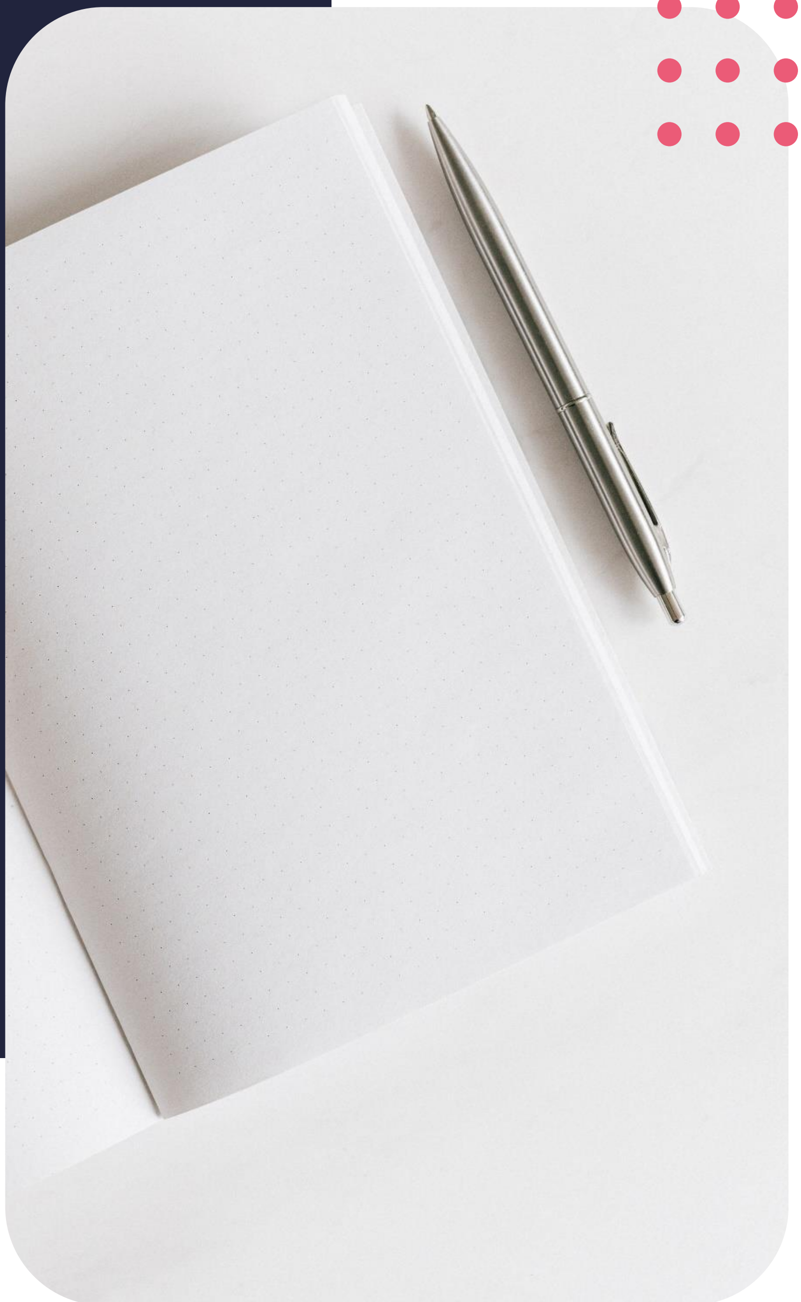

Lesson objectives

By the end of this lesson, you should be able to:

- Define project management
- Examine the project management lifecycle
- Unpack different project methodologies
- Identify the people involved in project management
- Highlight the tools of the trade

Introduction

Welcome to the project management course and congrats on taking the step to better understand the process. In no time we will have you thinking and interacting like a project management professional. Terms like scope, work, breakdown, structure, Agile, deliverables and a whole lot more will become part of your everyday vocabulary. You will look at project goals, teamwork, and time frames through new, enlightened eyes. Our first lesson is to unpack the term project management.

Fun Fact

Did you know that according to a survey put out by the Project Management Institute in June 2021, the global economy needs 25 million new projects professionals by 2030? This is due to economic growth, an increase in the number of jobs requiring project management-oriented skills and retirement rates. That means that to close the talent gap and keep up with the demand, it is estimated that 2.3 million project managers are needed to fill these roles every day. So, in a changing world, the role of project managers or changemakers will become more important.

Lesson objectives

We start each lesson with a roadmap for that session. The objectives for this lesson are to look at what exactly is project management, the project management lifecycle. We are also going to unpack the two most popular project methodologies like Waterfall and Agile. We will identify the people involved in projects and highlight some of the tools of the trade.

Define project management

The obvious place to start is to unpack the term project management. According to the Project Management Institute or PMI, the term project refers to any temporary endeavour with a definitive beginning and end. Note the word temporary because it has a defined start and end date. Also note the word endeavour, which basically means an attempt to do something. It is a specific set of operations designed to accomplish something unique, a singular goal.

Projects happen everywhere all the time, you get projects at work, such as new ad campaigns, employing new staff and managing clients at home, which involves renovating your kitchen or even just managing your housework in your neighbourhood, which involves roadworks or even block parties or community projects, as well as personal projects getting fit, being more social or even doing this very course. Stop for a minute to consider the projects you are currently managing in the various aspects of your life, think work, physical, spiritual, relationships, home life, social, health and finances.

The project management lifecycle

Whether you're working on a small project for your department or a large multi-departmental initiative with sweeping corporate implications and understanding of the project management lifecycle is essential. The project management lifecycle is usually broken down into five phases that take it from the beginning to the end.

- **Initiation** where you identify the need, problem or opportunity.
- Then you have to **plan** decide how you can based on things like the resources and budget you have available.
- Thirdly, **execution**. That's the time to get to work,
- **Monitoring**. Keeping things on track and adjusting if and when you need to in order to reach your goals.
- Finally, **closure** is where you provide the final deliverables and determine its success; what worked and what didn't to learn from the next time.

Kindly note that a project process can often go back and forth between the planning, executing, and monitoring phases.

Let's take an example of a project life cycle, such as building a bridge. The bridge project requirements are as follows: Funding has been supplied by the council for our raw materials, labour and the design of the bridge. The purpose of this bridge is to allow hikers to safely cross the river and access a nearby picnic site without disturbing the surrounding nature. Situated in a nature reserve, we've been told that summer is the most popular hiking season. This will allow us three weeks to construct the bridge.

If we look at project management in its most basic form, we will approach the bridge-building project by looking at what you aim to achieve.

The goal is to allow hikers to cross the river safely.

How do you plan to achieve it? You're going to build a secure bridge with sturdy railings.

How much time do you have in this case? It's three weeks

What will the cost be? We have limited council funding for this project.

Every project can be broken into tasks, deliverables, constraints and milestones – we will discuss these in a later lesson.

How will you apply the lifecycle of project management in this bridge-building project? During the initiation phase, you would decide if it is a feasible option for you to build this bridge.

Define the project and setup the expectations. The project is to build a bridge, and the expectation is that it will help hikers safely cross the river. Once you've established how you are going to do all this, you will then move on to planning.

In the planning phase, you would need to consider how long you have to do what needs to be done. In this case, it would involve measuring the river width and proposed length of the bridge, finding the people who are going to help you and the different raw materials, establishing the budget based on all of the above. The planning stage is vital to consider all aspects of your projects.

In the execute phase, this is where you start implementing the project plan that details every little step that you need to go through in order to complete your project. It is like your little black book of exactly what needs to be done and when it needs to be done. The bridge will be physically built during execution.

Monitor: you're not just going to build the bridge and then walk away hoping that everything's OK. Monitor the bridge by checking the measurements, the railings and following the process of making sure that it is safe. If it is, in which case we've achieved our aims and we have happy stakeholders. The council is happy with what we've done.

We would then move on to closing the project. What if the bridge that is built is not up to standard? Let's say, for argument's sake, that the wood we've chosen has been identified as wood that will decay quickly due to weather conditions. You would need to go back through the process to find a new type of wood to replace through monitoring. You identified that the wood will not hold. You then have to go back and plan how you're going to fix this problem and then build again or replace certain elements with the suitable raw material.

Once that is done, you would go back to the monitoring phase, once evaluated again, if satisfactory, only then will the project move onto the closure phase.

Your key takeaway here is that this process can repeat multiple times and you can also cause a process of repeating tasks and iteration. In the closing phase of our project, you would compile all the documents related to the project and possibly prepare a report. These would then be handed over to the council.

Project methodologies

A project management methodology is a set of principles and practises that guide you in organising your project to ensure their optimum performance. Agile and Waterfowl are two distinctive methodologies of processes to complete projects or work items. We're going to look at both of them right now.

Waterfall is the traditional approach. It is a sequential methodology. Tasks are generally handled in a more linear process. This is why it's called a Waterfall approach, because the water is all flowing down and can't go back up.

Agile is the more modern, flexible approach to project management. It is an iterative methodology that incorporates a cyclic and collaborative process.

Let's first look at the Waterfall methodology. As mentioned, it is a linear approach that relies on dependencies, meaning that tasks can't be done without the previous task having been completed, as we did with the lifecycle example of building a bridge. The best way to see the Waterfall methodology in action is to use an example. Let's say, designing a website for a client because the Waterfall method is often used in the I.T. industry. Start off by defining your requirements.

The key aspect of the Waterfall method is that all your requirements are gathered at the beginning of the project. This would allow you to then plan each phase carefully without involving the customer any further. The key here is to ensure that you've gathered every requirement because your customer will not see the project until it is complete. The design phase is broken up into logical design sub-phases, taking it bit by bit.

For the website design example, you would have to design the homepage first, then you would move on to the contact us page and then logically move through each phase of design. During this time, you will then brainstorm different ideas for how you'd like to design this website.

In the implementation phase, the programmers or web developers will start working on the functionality of the website, allowing end users to navigate through the website and perform different functionalities. Basically, it's bringing the design to life.

In the verification phase, the customers review the project to make sure that it meets all the requirements that were laid out in the requirements phase. This is when you would hand over the product to the customer for them to check. Remember, until this point, they are totally unsure of what you're going to give them.

The final phase is maintenance. Your customer will be using the software daily, so it is important to watch for any features that aren't working correctly, areas that might be showing up or even the speed of the programme. Technology is changing every day, so it is also vital that you monitor what is going on with your software and make sure that it is still relevant, especially if you'd like to be used for further projects in the future.

Let's now look at the Agile method using the same website example. We'll start off by planning our project, answering the questions, what do we want to achieve in this project? What is our project goal once our initial planning has been done?

We will then move on to defining the requirements. What exactly does the customer want and how do we plan to deliver it once we know all of this?

We will then move on to development and implementation. This is physically writing the code and making the website live so that we can see exactly what is going on and how everything is working.

Once the developers are confident in what they've done here, we will then move on to the initial testing. This is essentially when another team of people tries to break what you've made. This is where you'll pick up bugs like different web pages not working. We all know the frustration of being on a website when you can't navigate to the next page or even navigate back to where you were before. Initial testing is used to identify these errors or bugs within your website. Once you found all your bugs and errors, you'll then move on to developing and implementing again.

Once you've done that, those changes will then be tested, so your client will now test the website. This is a vital phase within the Agile process because there are two ways that this can go: If the client is happy, then you'll move on to system testing. And if everything goes well within your system testing, you will then move on to releasing the websites and making it live so that users and their potential customers can access this website, something that often happens within your system.

If the client is not happy and it doesn't go your way, you will move on to a second iteration of this Agile process within the client testing. Again, you will identify problems and changes that need to be made. Then you will find out how you are going to incorporate these changes. You'll then have to decide again whether the website is complete not.

There is no real limit to how many times you can go around the Agile methodology, which is why it is very popular with software development.

Here are the steps to creating and testing a minimal viable product or an MVP:

We first have to identify the key elements, then try to save time and resources, check whether the product is appealing to potential users, then find the trends to take advantage of and acquire a potential user base. So, in summary, the Agile methodology is an incremental approach, while the Waterfall method is a sequential approach. Flexibility as you go along is key in the Agile approach, while the Waterfall approach is more rigid. Agile is a collection of different projects, while the Waterfall method would focus on the completion of a single project.

Key players in project management

A project management methodology is a set of principles and practises that guide you in organising your projects to ensure their optimum performance. The key players in any project are the project executives, project stakeholders, project sponsors and the project team.

Project executives are workplace management that directly manages short- and long-term projects, communicating with upper management and monitoring project performance both internally and externally.

Project stakeholders are stakeholders that are invested in the project. They will be affected by your project at any point along the way. Stakeholder input can directly impact the outcome of a project.

A project sponsor is a person or group who provides resources and support for the project and is accountable for enabling success. A project sponsor is one level above a project manager.

The project team is staff resources, who together have all the needed skills and expertise to complete the project. The project teams are made up of dedicated staff and part time staff. Team member activities include providing their expert advice, completing deliverables and objectives, ensuring documents are updated and work with a team to make the project a success.

In the broadest sense, project managers or PMS are responsible for planning, organising, and directing the completion of specific projects for an organisation while ensuring these projects are on time, on budget and within scope. The project manager plays a key role in the process, reporting to the project stakeholders, or owners, the project sponsor, and the project team.

In case you may be feeling a bit confused about the role of the project sponsor and the project manager here is the difference: while the project manager is responsible for the day to day operations of the project, the product sponsor seeks to promote the project to keep it high on the priority list.

Brain break

If we say that a project is temporary because it has a defined beginning and end, it must also have a defined scope and defined resources. So how do you measure its success? Project success is determined by different factors that, when measured, define the success of the project.

Project managers need to define the success criteria after collaboration with the end customer. Being on the same page regarding project success criteria can eliminate the risk of project failures and improve the odds of project success.

Measurable outcomes

Three measurable factors that can define project success are: the scope, which is the amount of work required to complete the project. The time required or allowed for and the budget or costs.

These three come together to form what we call the iron triangle, or a triple constraint. In traditional project management, the iron triangle consists of three constraints: scope, which is fixed and refers to the features and functionality that the project is aiming to deliver. Time refers to the schedule and the amount of time it will take to complete the project. Cost refers to the amount of money required to pay for the resources, as in material equipment and tools and the salaries of people that work on the project. Both time and cost can vary. This model is also known as the triple constraint for obvious reasons. Basically, if you change one of the constraints, the other two will be affected.

This is how projects with big bundles of upfront requirements ran late or overran the budget. It was also known as the Iron Triangle in traditional project management, because the scope of projects was usually fixed costs in iron and unchanging, while the time and cost could fluctuate in order to meet the scope. For example, more time and an increase in the budget can be allowed for in order to ensure that the scope of the project is met.

In Agile project management, we flip the upside down. The schedule or time and resources costs are fixed while the scope varies. This is because in Agile projects, a fixed duration is selected for the iteration or sprint, and the Agile team remains the same.

Remember we mentioned that word earlier: iteration is a key term in project management. Iteration is the amount of time allocated to completing the work committed to for delivering functionality to the customer or end user and iteration is usually between two to four weeks long. The amount of scope that can be completed in each iteration is determined by the length of the iteration and the skills and experience of the team. This is the reason why prioritisation in Agile projects is so important.

If only a certain amount of scope can be accomplished in each iteration, the team needs to ensure that the highest priority requirements with the biggest return on investment are selected. This is also one of the biggest benefits of Agile in that it controls the work that is undertaken. Therefore, requirements that add no or little value are moved lower down the priority list.

Back to the iron triangle to explain it, using our bridge-building example we tackled at the start of this lesson: scope is the amount of work required in order to complete the project. In order to build the bridge, we need to do X number of activities, as we spoke about earlier, as in building the frame, laying down the wood, the cement foundation. All that would then fall into the scope time would be the fact that we only have three weeks in order to complete the project.

Cost would be the funds allocated to us by the council. All of these things together will determine the quality of the product that we produce or the quantity of the bridge that we produce. If suddenly you only had three days to complete the bridge. Well, that would be very different results to the bridge that you had three weeks. To prepare your own triangle, link all three of these things together in order to determine the quality of your product.

Another very important factor to remember is that your outcomes are measured also by your stakeholder satisfaction. You can deliver everything here perfectly, but if your stakeholders are unhappy, your project will not be considered a success. You can do the bridge within three days, get it done and do it within budget. But that doesn't necessarily mean that you have delivered the quality that was expected of you. You have to remember that your stakeholder satisfaction is key. There is that saying in project management, quality equals success.

Tools of the trade

Project management tools are the project managers' answer to manage projects. Simple projects require nothing more than a checklist, while other complex ones require proper planning, assigning tasks, setting deadlines, making sure that everyone sticks to them and tracking the time spent.

Some of the most important features in software project management tools include things like task management, team collaboration, learning materials, email, integration, document management and mobile capability.

There are so many software programmes available to help you to plan, organise and manage your team's work from start to finish. Here's a list of some of the more popular options.

- ProofHub
- Asana
- MicrosoftTeams
- Smartsheet
- Workzone
- Trello
- Monday.com

Please note that we will be using Monday.com for this course. It is a cloud-based platform that allows companies to create their own applications and work management software. It is simple but intuitive and can literally empower your team to do anything.

Lesson challenge

We would like to set you a challenge to complete before the next lesson. Do your own research on the role of the project manager and think of a situation at work or in a non-work setting where you can assume the role of a project manager. Jot down your duties and responsibilities to ensure the success of the project. In our next lesson, we are going to unpack the first phase of the project management cycle: the initiation phase. We'll talk about feasibility studies, stakeholders and how to create a project charter. So be sure to join us next time.

Conclusion

This concludes your first lesson in Module one for the course. Make sure you have schedule time for lesson two of Module one to the most convenient time for you. That's all for today. See you in the next lesson.

References

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